

Pathotype Prediction of *E. coli* Genomes Through Genomic Analysis and Machine Learning

Grace A. Driskill¹, Swapna J. Gadre¹, Ren-Chu Wang¹, and Dorian J. Feistel²

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Abstract

Understanding the genetic determinants of pathogenicity in *Escherichia coli* is pivotal for effective diagnosis and treatment strategies. Leveraging the power of whole-genome shotgun short-read sequencing, we propose integrating machine learning algorithms to enhance pathotype prediction, addressing current gaps in genomic analyses. Using a dataset of approximately 22,000 sequenced *E. coli* isolates, we identified pathotype-specific virulence genes using the BLAST algorithm, revealing six unique pathotypes and 427 unique virulence factors. Additionally, we employed Multiple Correspondence Analysis (MCA) to examine patterns and relationships among pathotypes based on their virulence factor profiles. Our results revealed distinct clusters of pathotypes, with Enteropathogenic *E. coli* (EPEC) and Enteroinvasive *E. coli* (EIEC) showing notable separation based on their virulence gene profiles. Furthermore, we applied Recursive Feature Elimination with Cross-Validation (RFECV) for feature selection, identifying 71 virulence genes that significantly contribute to pathotype classification. Evaluation of machine learning models trained on the selected features demonstrated improved performance metrics. Our findings provide valuable insights into the genomic landscape of pathogenic *E. coli* strains and highlight the

utility of computational approaches in deciphering virulence mechanisms. This study contributes to advancing our understanding of bacterial pathogenesis and informs the development of targeted interventions for mitigating the impact of *E. coli* related infections.

Introduction

Escherichia coli represents a diverse group of bacteria which naturally inhabit the intestinal tract of humans and animals (Escobar-Páramo et al., 2004; Gordon et al., 2015). While many *E. coli* strains are beneficial and form part of a healthy gut flora, acting as commensal (harmless) organisms, there are specific genetic variants that have evolved into significant pathogens (Mainil, 2013). These pathogenic strains can cause a range of illnesses, from mild diarrhea to severe infections, thus posing a major public health concern.

Pathogenic *E. coli* can be categorized into what is known as pathotypes based on their unique mechanisms of disease (Geurtsen et al., 2022). Several virulence factors—genes that enable a microbial pathogen like *E. coli* to colonize, infect, and/or evade immune defenses in its host organism—have been identified as pathotype specific associated genes, which can be used to classify the various *E. coli* pathogens into one of several pathotype backgrounds

¹ School of Computer Science and Engineering

² School of Biological Sciences (Bioinformatics)

(Denamur et al., 2021). For example, Diffusely adherent *E. coli* (DAEC) is recognized by the presence of the Afa/Dr adhesins genes (Servin, 2014); Enteroaggregative *E. coli* (EAEC) carries the aggR regulon (Morin et al., 2013), controlling a plasmid-encoded operon responsible for aggregation; Enteroinvasive *E. coli* (EIEC) contains the ipaH genes located on the pINV plasmid (Pasqua et al., 2017), aiding in host cell invasion; Enteropathogenic *E. coli* (EPEC) harbors the eae gene within the locus of enterocyte effacement (LEE) pathogenicity island (Mare et al., 2021), causing tissue damage; Enterotoxigenic *E. coli* (ETEC) is characterized by the heat-labile (lt) and heat-stable (st) enterotoxin genes (Turunen et al., 2020); and Shiga toxin-producing *E. coli* (STEC) possesses stx1 and stx2 genes responsible for toxin production (Haque et al., 2022).

Whole-genome shotgun short-read sequencing has become an invaluable tool for in silico identification of *E. coli* pathotypes. However, sequencing data can fall short when isolates are partially sequenced, resulting in incomplete or draft genomes that may lack crucial sequencing data. For instance, plasmids tied to an isolate's pathogenicity are often not as thoroughly sequenced as bacterial chromosomes (Juraschek et al., 2021), producing only partial plasmidic sequence data. Additionally, issues with sequencing biases such as low breadth/depth of coverage and uneven GC content (Gunasekera et al., 2021) can impede accurate pathotype classification in subsequent downstream bioinformatic analyses. Therefore, relying solely on a few pathotype specific associated genes may fail to capture an *E. coli* isolate's pathotype, highlighting the need for a comprehensive genomic analysis to identify if additional

diagnostic marker genes increase the likelihood of *E. coli* pathotype identification.

To help solve this issue, we suggest integrating machine learning (ML) algorithms into the identification and prediction of pathotype *E. coli* strains, presenting a unique approach in microbial genomics and public health. ML models can be trained to recognize distinctive patterns between virulence factors associated between *E. coli* pathotypes. For example, one published approach explored several ML models to predict an *E. coli* pathotype; however, their approach focused only on predicting the eae-positive Shiga toxin-producing *E. coli* pathotype (Vorimore et al., 2023). Therefore, there we see a significant opportunity to develop a more robust model that addresses the broader spectrum of *E. coli* pathotypes. Furthermore, using ML models for predicting *E. coli* pathotypes may address the challenges posed by incomplete genomes and sequencing biases by providing a robust framework for identifying critically associated virulence factors to improve pathotype classification.

In this paper, we test several machine learning models--against whole-genome shotgun short-read sequences to identify the most effective ML approach in predicting *E. coli* pathotypes. Briefly, we leveraged a comprehensive database of approximately 22,000 *E. coli* genomes freely available from the National Center for Biotechnology Information ([NCBI](<https://www.ncbi.nlm.nih.gov>)) and detected six pathotypes across 3,922 genomes and 1,379 genomes classified as non-pathotype. Virulence genes were detected across those genomes and a feature matrix was designed which consisted of 427 unique virulence genes. We then trained those ML classifier models on the raw data using a 70/30 split for training/validation

datasets to get baseline performance metrics (accuracy, precision, recall, F1-score) for each model. We then used two feature engineering techniques (Multiple Correspondence Analysis and Recursive Feature Elimination with Cross-Validation) followed by hyperparameter tuning to retrain the models using the same split ratio to determine if those would enhance the performance metrics. Finally, we test the models trained on the full dataset against unseen data from 190 draft (i.e. incomplete) *E. coli* genomes that were PCR and *in silico* confirmed to belong to a pathotype. In each section below, we discuss the methods and results together.

Methods and Results

E. coli Pathotype Determination

The predicted functional genes or nucleotide coding sequences (CDS) from approximately ~22k sequenced *E. coli* isolates were downloaded in fasta format from NCBI. Those CDS were used as query input into the BLAST algorithm for sequence alignment to detect the best matching homologous pathotype specific virulence genes (Table 1 [page 20]) at a minimum of 90% sequence identity and 50% sequence coverage (i.e. the query sequence length divided by the subject length multiplied by 100).

Out of the ~22k *E. coli* nucleotide assemblies downloaded from NCBI, 3,922 were unambiguously identified with a single pathotype. The majority of the pathotype identified genomes consisted of draft genomes at the Contig (95.49%, n=3745) assembly level, with the remaining genomes as Scaffold (2.67%, n=114), Complete (1.58%, n=62), and Chromosome (0.025%, n=1), with a mean depth of sequencing coverage of ~133X (Figure S1) and an

average of ~4976 predicted genes per *E. coli* genome (Figure S2). For negative controls, we screened 2,072 complete *E. coli* genomes against the same pathotype-specific virulence genes and found that 1,379 did not have any pathotype genes detected from Table 1. Given that these genomes are considered complete or 'closed,' we assumed the absence of these pathotype-specific genes was sufficient evidence to classify them as 'non-pathotype.' These were included in this dataset (data for coverage and gene count not included). The total number of genomes used in this analysis (before data cleaning) was 5,301 genomes.

Virulence Factor Profiling of Pathotypes

Next, we performed another BLAST sequence alignment with using the CDS from those 5301 genomes against 4,368 experimentally verified virulence factors (VF) downloaded from the Virulence Factor Database (VFDB) using the same minimal criteria as above. We detected a total of 427 unique VFs with a mean of ~89 VF genes per genome (Figure S3) suggesting that VFs only make up a small percentage (ranging between ~2.3% and 1.26%) of the total CDS across our pathotype identified genomes.

Feature Matrix Curation

We generated a binary matrix representing the presence or absence of those 427 VFs detected across all 5,301 pathotype and non-pathotype *E. coli* genomes and filtered the data in the following order: 1) removed features (genes) that had less than 10% variance (equal to 0.025 variance as the maximum variance for a feature in a binary matrix is 0.25) as suggested by Vorimore et al. (2023) (Figure S4), 2) removed pathotype-matching genomes with identical gene combinations, keeping only one genome as the representative, and 3) removed

any genes that were not found in any genome (in case the previous filtering creating features with all zeros). This produced a final feature matrix with 2411 genomes (samples) and 274 genes (features), producing a ~8.8:1 sample-to-feature ratio, which is close to the standard 10:1 sample-to-feature ratio. However, the percentage of genomes was imbalanced across pathotypes (Table 2).

Pathotype	Counts	Percentage
DAEC	71	2.94
EAEC	345	14.31
EIEC	200	8.29
EPEC	375	15.55
ETEC	79	3.28
Non-pathotype	823	34.13
STEC	518	21.49

Table 2: Distribution of pathotypes with corresponding counts and their percentage of the total (2411).

Methodology and Data Integrity in Model Testing

The models included in this evaluation were: Decision Tree (DT), Gradient Boosting (GB), K-nearest Neighbor (KNN), Logistic Regression (LR), Multi-layer Perceptron (MLP), Random Forest (RF), and Support Vector Classification (SVC). These models were chosen for their diverse approaches to handling data. For example, DT, GB, and RF can effectively handle nonlinear data and feature interactions through hierarchical, tree-based approaches; KNN relies on local data proximity for its predictions; LR models linear decision boundaries efficiently; SVC are optimal for creating non-linear boundaries using kernel tricks; and MLP are capable of learning complex and non-linear decision surfaces.

We tested the performance of each model by measuring accuracy, precision,

recall, and F1-score using two methods: 1) Train each model using 70% of the curated feature matrix while using the remaining 30% as validation, and 2) Train each model using the full curated feature matrix and using a test data set from data not included in the curated data (kindly provided by Dorian J. Feistel). Using these two methods, we were able to assess the internal consistency (overfitting) and external validity (generalizability) of our tested models, allowing us to verify the performance of each model under controlled, known conditions and evaluate their effectiveness in predicting outcomes on previously unseen biological data.

Our collection of *E. coli* genomes used as the unseen data set comprised of approximately 190 genomes. These were imbalanced in terms of their pathotype proportions and did not include all the pathotypes represented in the curated feature matrix. Furthermore, several virulence factors identified in the curated feature matrix were absent in the unseen *E. coli* genomes. We recognize this as a potential limitation to our study and anticipate that models trained on the full feature matrix and tested against the unseen data may exhibit more varied performance metrics compared to models trained on a significant portion of the training data and tested using the same curated data. Nevertheless, we believe our approach in using this data set provides a robust measurement that reflects real-world challenges within biological scenarios and offers valuable insights into the generalizability and practical applicability of determining which classifier model has the strongest predictive power.

Baseline Performance Metrics

When we used the curated feature matrix to train and validate each model, the

baseline accuracy score for most models was ~90% (range: 89-92%) with KNN having the lowest accuracy (77%) (Table 3). When we train the model on the full matrix and tested it against unseen data, accuracy scores were lower (Table 4). As mentioned earlier, we suspect that this reduction in accuracy is because of a few factors: 1) the unseen test dataset is much smaller which can make precision scores vary when calculating, 2) several pathotypes used to train the models were not present in the unseen test data, which can bring down scores if not properly handles in the calculations, 3) several features that were used in the training dataset were not present in the test data leading to natural biological challenges (i.e. the 'missing' genes may have not been detected due to sequencing bias/errors or because they genuinely do not exist in the *E. coli* genomes in the test data possibly reflecting spatiotemporal genetic variation)

Given the imbalance of pathotype genomes used in the training data (Table 2), we chose to also measure the weighted F1 score to provide a more comprehensive assessment of the predictive power of our tested models. Notably, the baseline F1 scores were higher across all models compared to the accuracy when trained and tested on the 70/30 curated feature matrix (Table 3) and when tested on unseen data (Table 4). Despite the higher F1 scores observed when training and testing the models using the curated feature matrix (Table 3), the models still performed well (except for KNN) with respect to their F1 scores when trained on the full feature matrix and tested with unseen data (Table 4), which demonstrates their robustness to generalize unseen data.

Model	Accuracy	Precision	Recall	F1 Score	Duration (sec)
RF	91.1	95.61	95.58	95.42	0.4
GB	90.46	96.21	96.41	96.2	6.22
LR	89.27	95.34	95.44	95.3	38.27
KNN	77.83	86.12	86.05	85.71	1.13
MLP	89.26	95.84	95.86	95.72	713.61
SVM	83.47	90.93	93.09	91.8	0.57
DT	92.06	95.46	95.3	95.36	0.12

Table 3: Baseline performance metrics when using 70/30 testing/validation to train/validate all models using the full curated feature matrix.

Model	Accuracy	Precision	Recall	F1 Score	Duration (s)
RF	83.01	100	80.63	88.52	0.5
GB	79.45	98.89	82.72	87.97	8.09
LR	77.02	99.41	74.87	83.29	37.39
KNN	74.28	95.37	63.35	72.08	2.23
MLP	84.09	100	81.15	88.92	796.79
SVM	79.25	100	81.68	87.81	0.64
DT	81.8	96.72	81.68	87.21	0.14

Table 4: Baseline performance metrics when training all models on the complete feature matrix and testing against unseen data.

Pathotype Virulence Factor Profiles

We employed a Multiple Correspondence Analysis (MCA) to explore patterns and relationships within and between pathotypes. The MCA plot ([Figure 1](#)) shows the spatial distribution of various *E. coli* pathotypes based on their VF profiles across the first two components, which collectively explains 35.85% of the variance. The distinct clusters of EPEC (blue) and EIEC (green) pathotypes show the most considerable separation across the first two components which suggest that there are specific genes that are strongly associated with these groups indicating a unique VF profile.

In contrast, the other pathotypes such as STEC (red), DAEC (yellow), EAEC (orange), and ETEC (purple) are mostly interspersed (except for a few STEC genomes), particularly along Component 2, suggesting that while they have some distinguishing features, there is a higher degree of similarity between their full VF profiles that define these pathotypes. Furthermore, the non-pathotype (grey) are notably scattered throughout the center of the plot, overlapping with STEC, DAEC, EAEC, and EAEC pathotypes to varying degrees,

indicating that non-pathotype identified *E. coli* share a similar VF profile with the other pathogenic strains.

Additionally, we performed agglomerative (bottom-up) clustering using a Jaccard distance and weighted linkage method to create a cluster map of the VF profiles between pathotypes. This allowed us to observe how well each pathotype (or non-pathotype) class clustered together based on their VF profiles (Figure 2). EIEC (green) and EPEC (blue) *E. coli* pathotype genomes clustered exclusively together based on a distinct set of VF genes, while the remaining genomes formed smaller pathotype clusters that making it unclear how their VF profiles influenced its pathotype (Figure 2).

The clear divergence of EPEC and EIEC pathotypes on first two components (Figure 1) as well as the cluster map (Figure 2) highlight differences in VF profiles between these groups. However, the overlap observed among DAEC, EAEC, ETEC, STEC, and non-Pathotypes, suggests that their VF profiles are less distinctive.

Dimensionality Reduction using MCA

After our initial assessment of all features (n=274) using the MCA (Figure 1) we plotted the explained percent variance (blue; left y-axis) and the cumulative explained variance percent (red; right y-axis) across all component and found that the first 40 components explained ~90% of the variance (Figure 3). Using this as a guide, we retrained all the models using the reduced MCA data (i.e. first 40 components) using the same two methods mentioned above and generated precision metrics to see how well they compared to the baseline results (Tables 5 & 6).

Model	Accuracy	Precision	Recall	F1 Score	Duration (s)
RF	85.33	92.16	92.27	91.82	1.07
GB	85.67	92.96	92.96	92.6	19.33
LR	86.6	94.35	94.06	93.43	0.06
KNN	83.45	89.07	88.95	88.59	2.44
MLP	88.63	94.8	95.03	94.82	1.56
SVM	84.38	94.03	92.96	91.97	0.33
DT	79.69	84.24	84.39	84.26	0.14

Table 5: Performance metrics when using 70/30 testing/validation to train/validate all models using only first 40 components after MCA transformation.

Model	Accuracy	Precision	Recall	F1 Score	Duration (s)
RF	70.21	100	68.59	76.72	1.46
GB	77.48	97.98	69.11	79.23	28.07
LR	81.25	100	78.01	86.67	0.09
KNN	80.74	100	81.15	87.47	1.72
MLP	84.52	98.36	82.72	89.5	2.54
SVM	74.6	86.91	80.1	83.22	0.31
DT	73.15	87.37	72.25	78	0.2

Table 6: Performance metrics when training all models with the first 40 components after MCA transformation and testing MCA transformed unseen data

Feature Selection using Recursive Feature Elimination with Cross-Validation

We first attempted to use L1 regularization with Logistic Regression to get features that had a correlation coefficient greater than zero. However, the precision metrics for all trained models using the 70/30 training data were significantly lower than the baseline results (data not shown), and thus we did not include these data in any further analysis. Next, we used a method known as Recursive Feature Elimination with Cross-Validation (RFECV), which is a feature selection technique that recursively removes less informative features based on their impact it has on model performance using cross-validation (n=5). The RFECV function relies on using an estimator to select the subset of features. We chose to use Logistic Regression as the estimator to curate a reduced feature matrix due to interpretability of feature correlation coefficients and computational efficiency. The clear interpretability of feature coefficients helps to reveal relationships between features and target variables, while

the model's simplicity enhances its computational speed.

This technique selected 71 features which each correspond to a virulence gene. The coefficients in the LR model used by the RFECV allowed us to gain insight on the relationship between specific VFs and pathotypes. Positive coefficients suggest a positive association between a gene and a particular pathotype, while negative coefficients suggest a negative association. The maximum and minimum coefficient values for each pathotype class are shown in Table 7.

Pathotype	Max/Min Coefficient	Max/Min Coefficient Gene/Function
DAEC	(1.446, -1.661)	(draC Adherence, sfaC Adherence)
EAE	(2.068, -1.021)	(aap Others, gspC Effector)
EIEC	(1.858, -1.019)	(ipaH Effector, csgA Adherence)
EPEC	(1.671, -0.605)	(cesD2 Effector, espX4 Effector)
ETEC	(1.715, -1.12)	(paa Adherence, stx2A Effector)
Non-pathotype	(1.771, -2.542)	(sfaC Adherence, stx2A Effector)
STEC	(4.803, -0.476)	(stx2A Exotoxin, espL1 Effector)

Table 7: Maximum and minimum feature coefficients and corresponding gene during RFECV feature selection.

In training/validation metrics, nearly all models had an increase in both balanced accuracy and F1 score compared to the baseline model results. The exceptions to this were GB, which had a decrease in accuracy and F1 score, and MLP, which had a decrease in F1 score. The accuracy averaged for all models did not change compared to the baseline, both ~90%. The average F1 score did increase from ~93.6% in the baseline to ~95.1% using RFECV feature selection (Table 3, 8).

When testing on unseen data, all models produced saw a decrease in the performance metrics except precision (Table 8 & 9). This is due to factors previously discussed, including the small size, and missing pathotypes and virulence factor in the test data. When compared to the baseline results on unseen data, only RB, GB and DT saw an increase in both balanced accuracy and F1 score. However, the average across all

models increases from ~79.8% to ~80.1% accuracy and ~85.1% to ~86.5% F1 score (Table 4, 9).

Another result to point out is that KNN had the lowest performance in accuracy and F1 by over almost 3% compared to the other models in the baseline (Table 3, 4). After feature selection, KNN was no longer the obvious worst model. While KNN still had the lowest F1 scores (92.81% for training/validation; 81.41% for testing), this was less than 1.5% off from the next lowest score. Additionally, SVM had the lowest accuracy in the training/validation and testing (Table 8, 9).

Model	Accuracy	Precision	Recall	F1 Score
RF	92.7	96.25	96.27	96.24
GB	90.08	96.24	96.13	96.1
LR	91.85	96.94	96.82	96.65
KNN	86.94	93.26	92.96	92.81
MLP	91.51	95.72	95.3	95.47
SVM	86.59	94.87	94.61	93.98
DT	91.21	95.07	94.2	94.54

Table 8: Performance metrics when using a 70/30 testing/validation to train/validate all models using only the reduced feature matrix

Model	Accuracy	Precision	Recall	F1 Score
RF	85.83	100	87.43	92.38
GB	83.24	99.52	84.82	89.99
LR	76.22	100	74.35	82.89
KNN	76.48	98.99	70.68	81.41
MLP	79.99	99.52	76.44	84.9
SVM	76.37	100	79.06	85.1
DT	81.96	100	83.25	88.81

Table 9: Performance metrics when training all models on the reduced feature matrix and testing against unseen data

Hyperparameter Tuning

We perform hyperparameter tuning by employing a grid search scheme on the hyperparameter space. In grid search, we define a set of hyperparameter values for each hyperparameter, the algorithm then

evaluates all possible combinations of these values. The grid search procedure works as follows:

1. Define a grid of hyperparameter values (for example, learning rate for MLP, regularization strength for regression models, and number of trees for random forest).
2. For each combination of hyper parameters, we train the model and evaluate its

Model	Dataset	Precision	Accuracy	F1 Score	Recall	BestParameters
Decision Tree	MCA	1	1	1	1	{'max_depth': 1, 'splitter': 'best', 'criterion': 'entropy'}
SVC	MCA	1	1	1	1	{'C': 1, 'kernel': 'poly', 'max_iter': 10}
Gradient Boosting	MCA	0.998631	0.998619	0.99862	0.998619	{'max_depth': 3, 'n_estimators': 100, 'learning_rate': 0.1, 'criterion': 'friedman_mse'}
Logistic Regression	MCA	0.997356	0.997238	0.997244	0.997238	{'C': 1, 'penalty': 'l2', 'max_iter': 100, 'random_state': 0}
KNN	MCA	0.993157	0.993094	0.993104	0.993094	{'n_neighbors': 1, 'algorithm': 'auto'}
MLP	MCA	0.990976	0.990331	0.990479	0.990331	{'max_iter': 100, 'learning_rate_init': 0.001}
Random Forest	MCA	0.973366	0.972376	0.971371	0.972376	{'n_estimators': 200, 'criterion': 'gini', 'max_features': 'sqrt', 'max_depth': ""}
Logistic Regression	RFECV	0.966986	0.966851	0.965558	0.966851	{'C': 1, 'penalty': 'l2', 'max_iter': 100, 'random_state': 0}
Gradient Boosting	RFECV	0.965743	0.96547	0.965183	0.96547	{'max_depth': 3, 'n_estimators': 400, 'learning_rate': 0.1, 'criterion': 'friedman_mse'}
SVC	RFECV	0.965808	0.96547	0.964516	0.96547	{'C': 1, 'kernel': "linear", "max_iter": 100}
Gradient Boosting	Raw	0.965684	0.96547	0.964273	0.96547	{'max_depth': 3, 'n_estimators': 100, 'learning_rate': 0.1, 'criterion': 'friedman_mse'}
MLP	RFECV	0.962319	0.959945	0.960764	0.959945	{'max_iter': 100, 'learning_rate_init': 0.01}
SVC	Raw	0.962288	0.961326	0.95964	0.961326	{'C': 10, 'kernel': 'rbf', 'max_iter': 1000}
Random Forest	RFECV	0.960315	0.961326	0.959006	0.961326	{'n_estimators': 400, 'criterion': 'entropy', 'max_features': 'sqrt', 'max_depth': ""}
Decision Tree	RFECV	0.958505	0.955801	0.956816	0.955801	{'max_depth': "", 'splitter': 'random', 'criterion': 'entropy'}
MLP	Raw	0.955336	0.955801	0.954706	0.955801	{'max_iter': 100, 'learning_rate_init': 0.01}
Logistic Regression	Raw	0.954843	0.95442	0.954078	0.95442	{'C': 1, 'penalty': 'l2', 'max_iter': 100, 'random_state': 0}
Random Forest	Raw	0.956776	0.955801	0.952478	0.955801	{'n_estimators': 100, 'criterion': 'gini', 'max_features': 'sqrt', 'max_depth': ""}
Decision Tree	Raw	0.946742	0.944751	0.945562	0.944751	{'max_depth': "", 'splitter': 'best', 'criterion': 'entropy'}
KNN	RFECV	0.925416	0.924033	0.924183	0.924033	{'n_neighbors': 1, 'algorithm': 'auto'}
KNN	Raw	0.897334	0.895028	0.895236	0.895028	{'n_neighbors': 1, 'algorithm': 'auto'}

Table 10: Best models after performing a grid search sorted by descending F1-scores

performance on a couple of metrics such as F1, precision, recall, and accuracy.

3. Select the best performing parameters on the evaluation.

The grid search scheme has the benefits of performing exhausting search over the entire hyperparameter space, so that no combination of hyperparameter is missed, and that it is very simple to implement. However, grid search is very computationally intensive because it scales linearly to the product of the number of configuration and might not find the optimal solution if the grid is too coarse.

The hyperparameter tuning is performed on the standard task, as well as the MCA task and L1 task. The parameters alongside their performance are stored with in a table format, reversely ordered by the F1 score, and available in the project release for future reference. Then, we trained the model using the fully curated matrix and present the evaluation results on the testing set (Table 10).

Our engineered features vastly outperform the raw features. We are able to consistently observe a 5% improvement in precision for our dimension reduction (MCA) dataset. The feature reduction (RFECV) is less effective compared to MCA, but still be able to deliver noticeable improvement in precision. We also noticed that some the hyperparameters themselves are quite robust against feature engineering techniques. For example, for SVC, parameter ``max_iter`` ends up being ``10`` for the best performing model over all instances. The same pattern is observed in decision tree's ``criterion``, KNN's ``algorithm`` and ``n_neighbors`` etc. This pattern suggests that these parameters might be the best for this particular problem size and distribution.

Discussion

In this study, we leveraged machine learning algorithms to predict *E. coli* pathotypes based in whole-genome shotgun short-read sequencing data. Our analysis focused on identifying pathotype specific virulence factors and evaluating performance of various ML models in pathotype classification.

We first performed a comprehensive analysis of approximately 22,000 sequenced *E. coli* isolates obtained from the NCBI dataset. We detected 6 pathotypes within the subset of genomes, while the remaining isolates were classified as non-pathotype. Subsequently, we identified 427 unique virulence genes associated with these pathotypes, resulting in a feature matrix comprising of 7 classes and 427 features.

We evaluated the performance of seven ML models, including Decision Tree (DT), Gradient Boosting (GB), K-nearest Neighbor (KNN), Logistic Regression (LR), Multi-layer Perceptron (MLP), Random Forest (RF), and Support Vector Classification (SVC). Initial training and validation on a curated feature matrix yielded high accuracy scores (~90%) for most models, with Decision Tree performing the best. However, when tested against unseen data, accuracy scores decreased, highlighting the challenges of generalization to new biological datasets.

To address the high dimensional nature of the feature space, we employed L1 regularization and Multiple Correspondence Analysis (MCA) to reduce dimensionality while preserving the variance in the data. Retraining models using the first 40 components after MCA transformation improved or maintained metrics, indicating

the effectiveness of feature reduction in model generalization.

While dimensionality reduction techniques like MCA try to preserve the original structure and relationships within the data, feature selection techniques identify the most informative features in a dataset to enhance predictive accuracy. In our study, we employed RFECV (Recursive Feature Elimination with Cross-Validation) to select the most informative features. RFECV systematically removed less informative features based on their impact on model performance, resulting in a subset of 71 features corresponding to virulence genes. Logistic regression was utilized as the underlying estimator for RFECV, allowing us to gain insights into the association between specific genes and pathotypes through coefficient analysis. The coefficients derived revealed both positive and negative associations between virulence genes and pathotypes. For example, the DEAC pathotype has high positive correlation with the *draC* Adherence gene, indicating its potential role in pathogenicity. While the *sfaC* Adherence gene has strong negative correlation, suggesting a potential protective role against the pathotype.

After feature selection, nearly all models demonstrated improved performance in terms of balanced accuracy and F1 score compared to the baseline results. Notably, KNN, initially performing poorly in the baseline, showed a significant improvement in performance after feature selection. However, SVC exhibited the lowest accuracy in both training/validation and testing phases, indicating potential limitations in its applicability to our dataset. Hyperparameter tuning further optimized model performance by systematically exploring the hyperparameter space and identifying the best combination of hyperparameters for each model.

Our study provides valuable insights into the genetic determinants of pathogenicity in *E. coli* and contributes to the development of predictive models for classifying pathotypes based on virulence gene profiles. Future research directions may involve experimental validation of the identified genes to elucidate their roles in *E. coli* pathogenesis and explore their potential as therapeutic targets or diagnostic markers.

Project Contribution and Integrity Acknowledgment

We, the undersigned members of Group Project 26, hereby acknowledge and confirm that the summary of contributions outlined below accurately reflects each individual's efforts and roles within the project. Each team member has contributed original work and has accurately reported results, ensuring no falsification or fabrication of data while upholding the highest standards of academic integrity and transparency. By signing this document (abbreviation, data, time) below, we affirm the accuracy of these statements and express our mutual respect and recognition for the contributions made by each team member.

Grace A. Driskill: GAD, April 29th 2024, 17:57

Swapna J. Gadre: SJG, April 29th 2024, 19:50

Ren-Chu Wang: RCW, April 29th 2024, 19:03

Dorian J. Feistel: DJF, April 29th 2024, 16:50

Authors Contributions

Grace A. Driskill = GAD

Swapna J. Gadre = SJG

Ren-Chu Wang = RCW

Dorian J. Feistel = DJF

Coding = C

Writing = W

Conceptualization: DJF

Proposal: DJF (W) (Edits: GAD, SJG, RCW)

Data Curation: DJF

Visualizations: DJF

Contributions by sections:

1. Abstract: SJG (W)
2. Introduction: DJF (W)
3. *E. coli* Pathotype Determination: DJF (C/W)
4. Virulence Factor Profiling of Pathotypes: DJF (C/W)
5. Feature Matrix Curation: DJF (C/W)
6. Methodology and Data Integrity in Model Testing: DJF (W)
7. Baseline Performance Metrics: GAD (C), SJG (C), DJF (W)
8. Pathotype Virulence Factor Profiles: DJF (C/W)
9. Dimensionality Reduction using MCA: DJF (C/W)
10. Feature Selection using RFECV: GAD (W/C)
11. Hyperparameter Tuning: RCW (C/W)
12. Discussion: SJG (W)
13. GitHub Contributions: GAD (W/C), SJG (W/C), RCW (W/C), DJF (W)

The source code for this project is at: https://github.gatech.edu/gdriskill3/ML_CB_project

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Figure 1

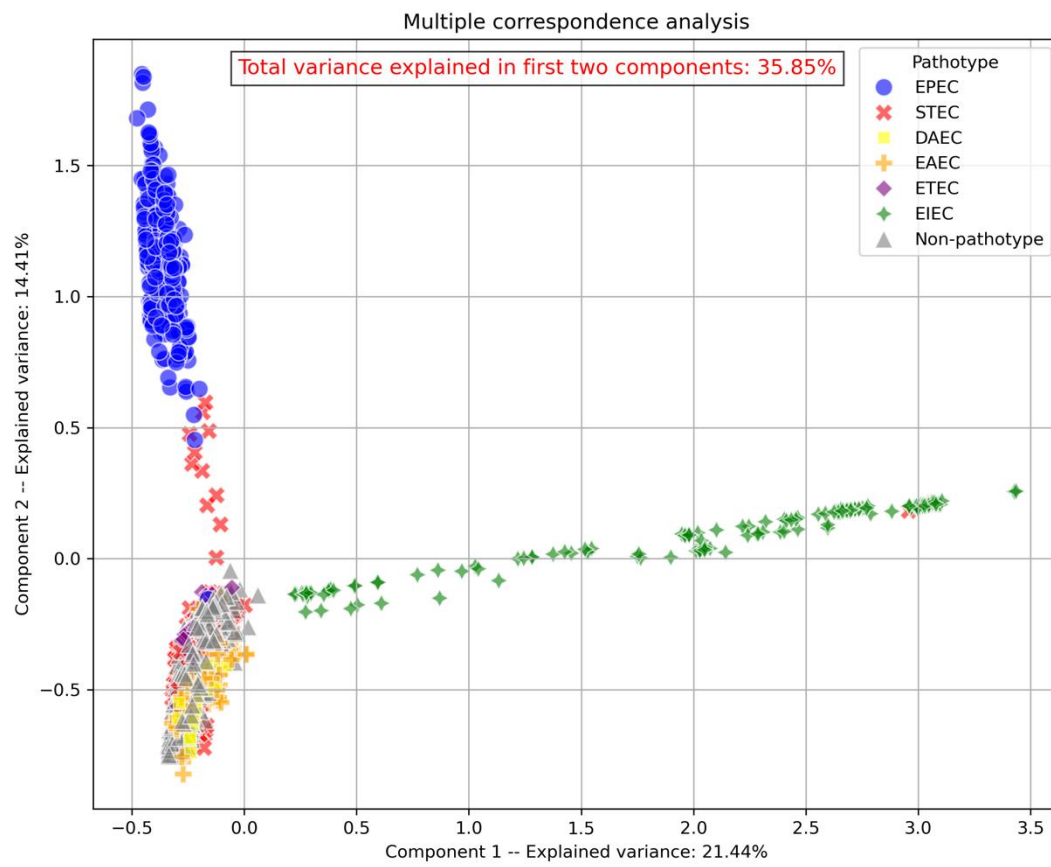


Figure 2

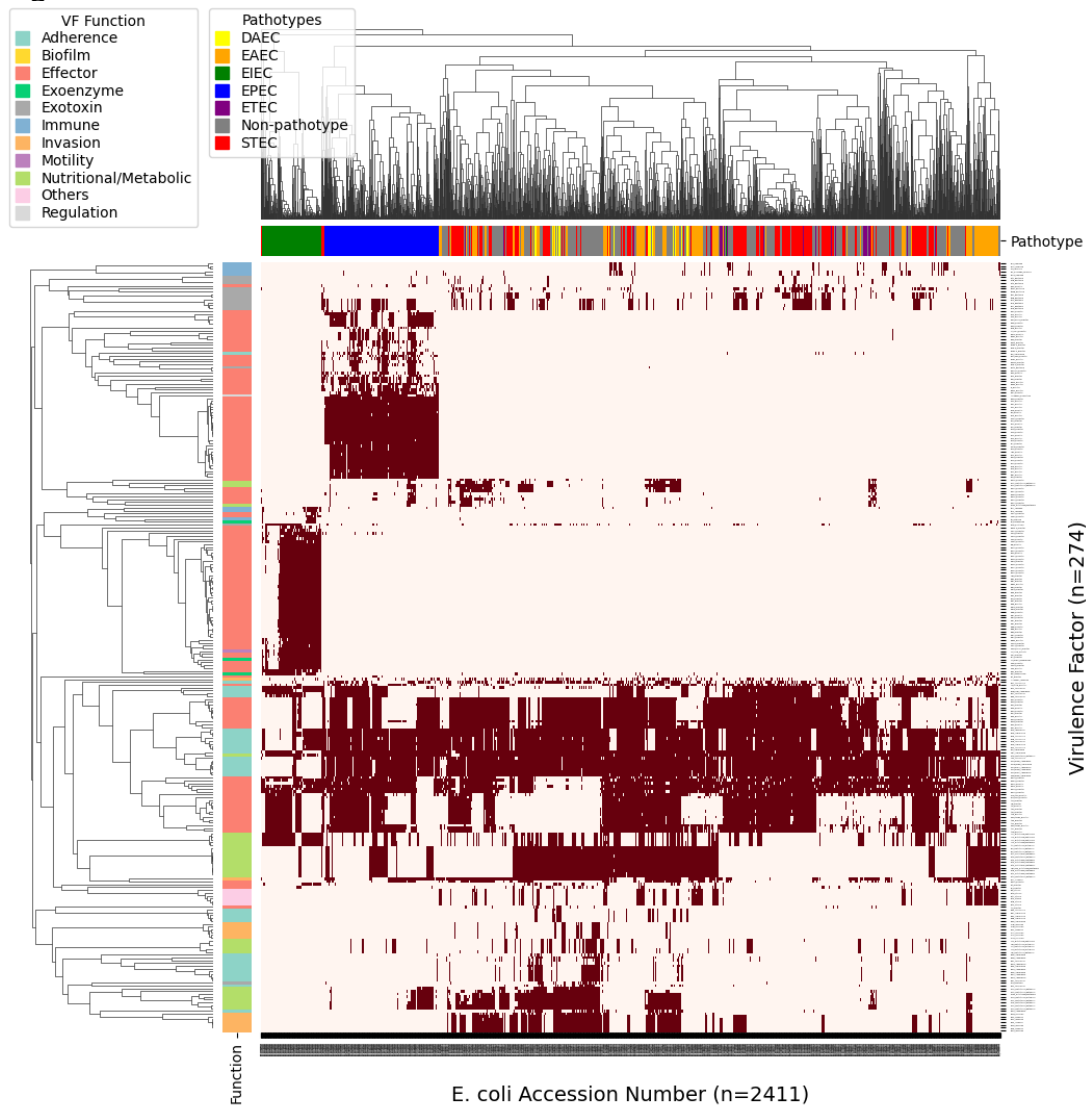


Figure 3

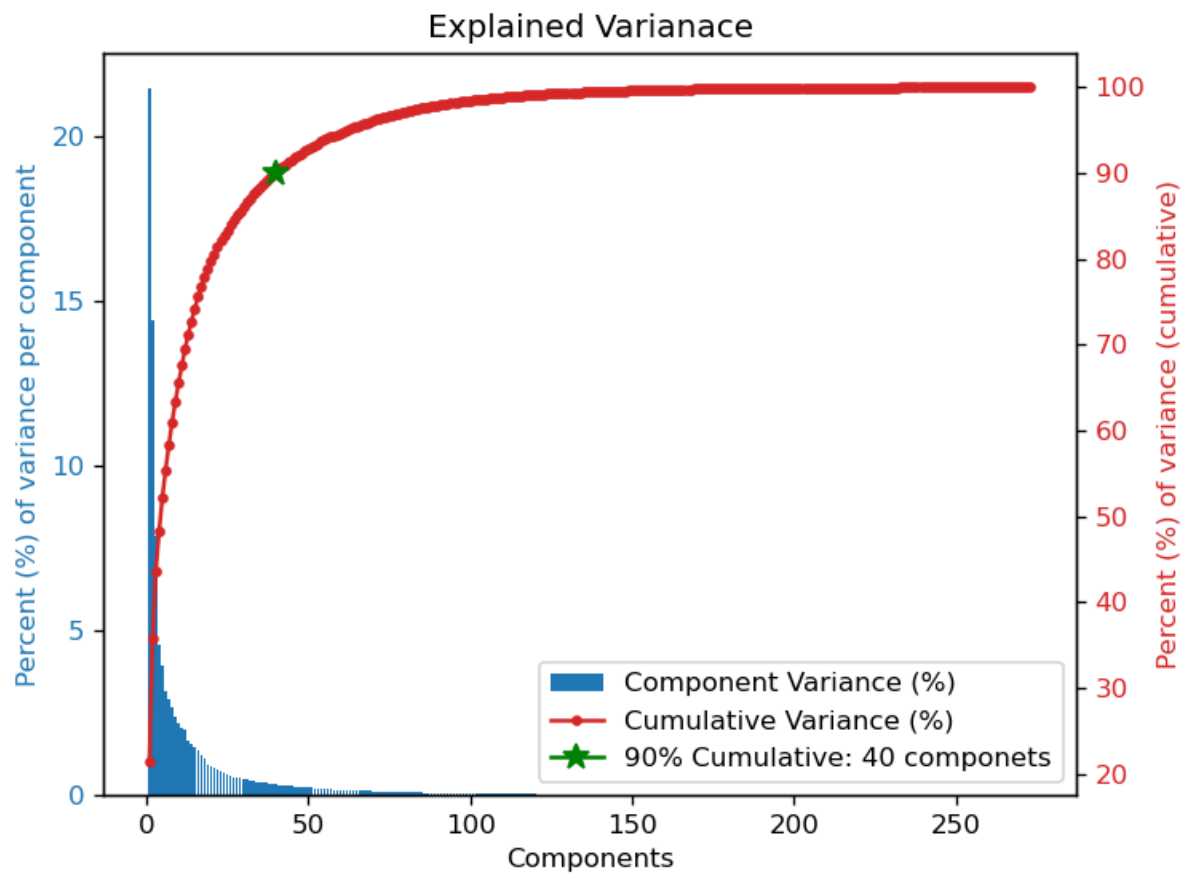


Figure S1

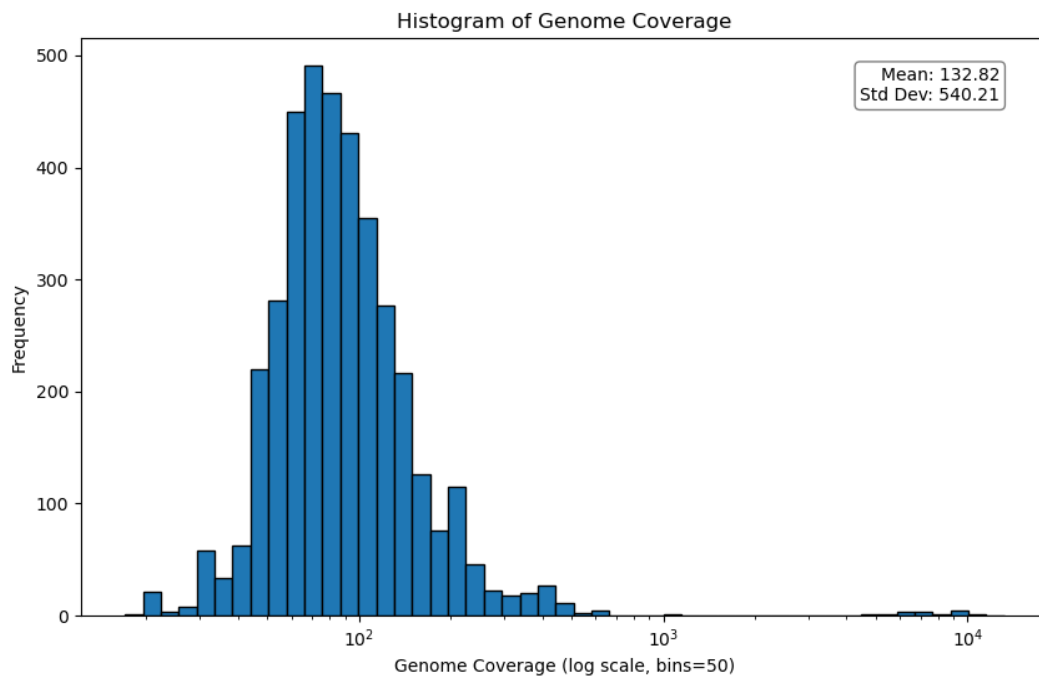


Figure S2

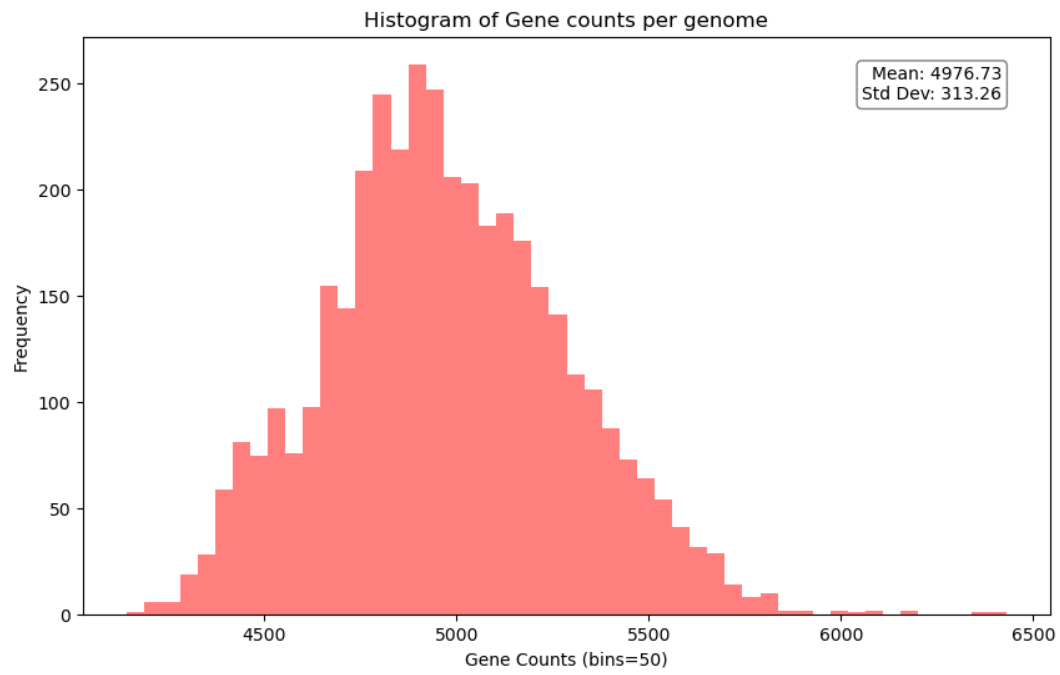


Figure S3

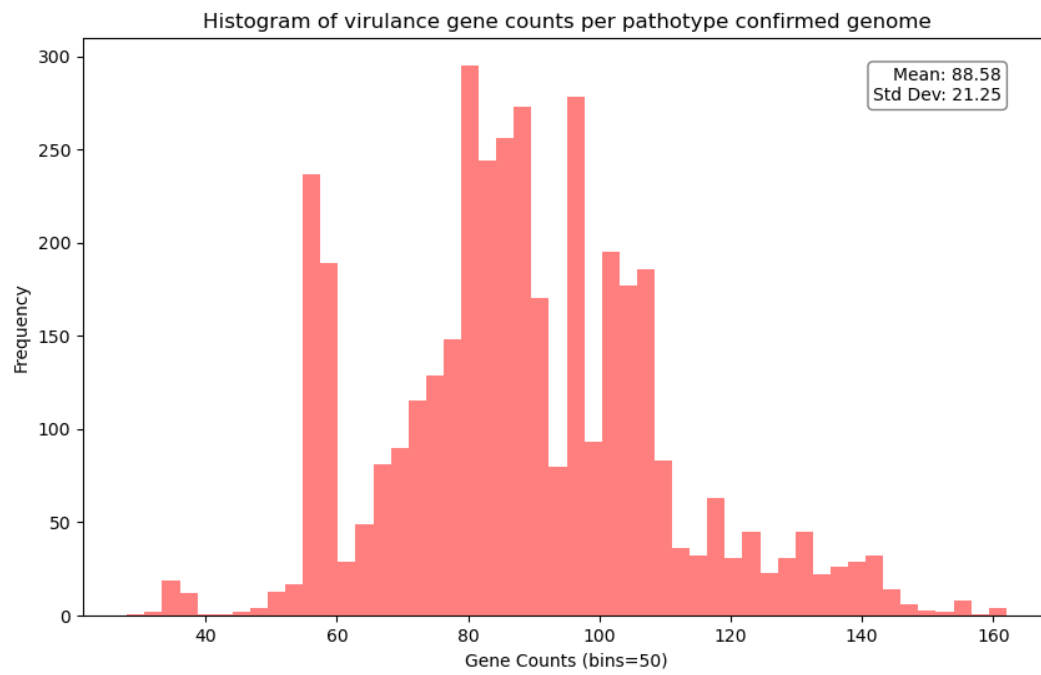
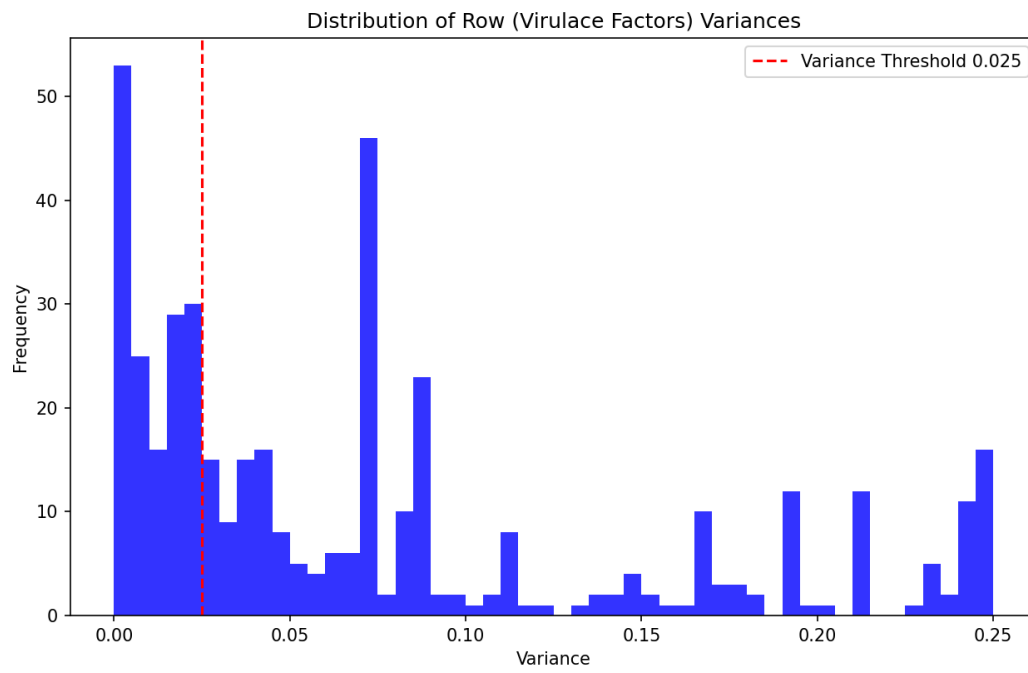


Figure S4



Pathotype	Pathotype designation criteria	Gene name	Gene description	Genbank Accession
ETEC	Isolates were designated as ETEC if either or both the <i>lt</i> (both subunits) or <i>sta</i> genes were present	<i>eltA</i> <i>eltB</i>	Heat-labile enterotoxin subunits A and B	AAA24685 AAA98064
EPEC	tEPEC is defined by the presence of both <i>eaeA</i> and <i>bfpA</i> ; aEPEC is defined by the presence of <i>eaeA</i> and absence of <i>bfpA</i> . For this analysis, we grouped these subpathotypes together as long as either gene was present	<i>eaeA</i> <i>bfpA</i>	Locus of enterocyte effacement-encoded intimin protein Bundle-forming pilus	AAC38392 BAA84838
EAEC	Isolates were defined as EAEC if <i>aggR</i> , <i>aaiC</i> , and/or <i>aatA</i> were present	<i>aggR</i> <i>aaiC</i> <i>aatA</i>	Aggregative adherence transcriptional regulator Secreted protein Biofilm formation protein	QKN22446 LT719075 SJK83517
DAEC	Isolates were considered DAEC if their draft genome sequences contained at least one of the two genes	<i>afaE-I</i> <i>afaE-III</i>	Afimbrial adhesin subunit E variant Afimbrial adhesin subunit E variant	CAW30801 CAA54121
EIEC	<i>E. coli</i> isolates with <i>ipaH</i> were designated as EIEC	<i>ipaH</i>	pINV plasmid-encoded type-III effector protein	NP_858212
EHEC	Isolates that had either <i>stx</i> gene designated as STEC	<i>stx1 A</i> <i>stx1 B</i>	Shiga-like toxin, subunits A and B	WP_000691354 WP_000752026
Table 1: Gene reference sequences used for each pathotype; gene presence was confirmed for isolate genomes with gene sequencing 90% sequence identity and 50% sequencing breadth coverage				